June 6, 1967
SUPERSEDING
Fed. Spec. O-H-202
October 15, 1964

FEDERAL SPECIFICATION

HERBICIDE, CHLORATE-BORATE MIXTURE

(CBM)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of a nonselective herbicide known as CBM, chlorate-borate mixture which combines the rapid plant destroying powers of sodium chlorate with the more persistent toxicity of sodium borate. It is a fast-acting weed killer affecting all types of vegetation and may be applied as a dry powder or in a water solution spray.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378 - Plastic Sheet and Strip (Polyolefin).
TT-I-558 - Ink, Marking Stencil, Opague for Nonporous Surfaces (Metals, Glass, etc.).
UU-S-48 - Sacks, Shipping, Paper.
PPP-B-35 - Bags, Shipping, Burlap, Cotton and Waterproof Laminated.
PPP-D-723 - Drums, Fiber.
PPP-P-704 - Pails, Metal: (Shipping, Steel, 1 through 12 Gallon).

Federal Standard:

Fed. Std. No. 123 - Marking for domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specification:

MIL-A-3941 - Adhesive, Paper Label, Water-Resistant.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Code of Federal Regulations:

7 CFR 362 - Agriculture, Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act.

(Copies of the above part of the Code of Federal Regulations may be obtained from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402)

3. REQUIREMENTS

3.1 Composition. The herbicide shall consist of the following ingredients in the proportions specified when tested in accordance with 4.5.1.

Ingredients	Percent by weight
Disodium octaborate tetrahydrate ($Na_2B_8O_{13}4H_2O$) (min.)	73.0
Sodium chlorate ($NaClO_3$) (min.)	25.0
Inert ingredients, including spreading agent (max.)	2.0

3.2 Solubility. The herbicide shall, when dissolved to a concentration of 1.75 lbs./gal. at 70° F., show not more than 0.1 percent insoluble material when tested as specified in 4.5.2.

3.3 pH. The herbicide shall have a pH of 7.0± 0.2 at 23° C. when tested as specified in 4.5.3.

3.4 Appearance and pour. The herbicide shall be a free flowing powder which shall be dry, loose, free from lumps, and foreign matter when tested as specified in 4.5.4.

3.5 Workmanship. The finished product shall be free from any defect which might impair its ability. It shall conform to all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot.

4.2.1 Herbicide. For purposes of inspection, a lot shall consist of all herbicides of the same concentration produced by one manufacturer under essentially the same conditions and process, within the following limitation. In the event the herbicide is produced by a continous-run process the lot shall contain herbicide produced in a period of not more than 24 hours. In the event the herbicide is produced by a batch process, each batch shall constitute a lot. Herbicide in the inspection lot shall be identified by order of production when continous-run process is involved and by batch number when a batch process is involved.

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4.2.2 Filled containers. For purposes of inspection, a lot shall consist of all specified containers filled with approved herbicide as required and which are to be offered for delivery at one time under a contract or order.

4.3 Sampling.

4.3.1 Sampling for test of herbicides. Three separate one-pound samples shall be taken from each inspection lot. When herbicide is produced by a continous-run process, the three samples shall be taken so as to represent, respectively, the first part, the middle part, and the last part of the run which constitutes the inspection lot. When herbicide is produced by a batch process, the three samples shall be taken at different locations within each batch.

4.3.2 Sampling for examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105, at inspection level I, and an AQL of 2.5 percent defective.

4.4 Examination.

4.4.1 Filled containers. Each sample filled container selected in accordance with 4.3.2 shall be examined for any defects of the container and the container closure, for evidence of leakage, and for illegible, incorrect, or missing marking. Each sample filled container shall also be weighed to determine the net weight of contents. Any filled container having one or more defects shall be considered a defective unit.

4.5 Tests. Samples of herbicide selected in accordance with 4.3.1 shall be combined and subjected to the tests specified herein. If the composite sample fails to pass any one of the tests specified, the lot represented therein shall be rejected. Rejected lots may be reinspected provided the manufacturer has corrected all defects. The manufacturer shall furnish to the Government full particulars concerning the rejection and the action taken to correct the defects.

4.5.1 Composition.

4.5.1.1 Determination of disodium octaborate tetrahydrate.

4.5.1.1.1 Reagents:

Mannitol, hexahydric alcohol, reagent grade.
0.5N NaOH
0.5N HCl
Methyl red - 0.10 percent alcohol solution.
Methylene blue - - - - - - - - - - - - - - - - - - - 0.25 percent water solution.
Phenolphthalein - - - - - - - - - - - - - - - - - - - 1.00 percent alcohol solution.

4.5.1.1.2 Procedure. Dissolve 1.000 ± 0.001 g. sample in 150ml. distilled water in a 500 ml. Erlenmeyer flask. Add 5 drops of methyl red and acidify with 0.5N HCl from a buret, adding 0.5 ml. excess. Add a few glass beads to the flask, cover with a watchglass, and boil 5 minutes to remove CO_2; cool in running water. Add 1 drop methylene blue and back titrate excess 0.5N HCl with 0.5N NaOH to a bluish-green end point. Calculate the net 0.5N HCl required to neutralize the sodium oxide. Adjust the buret reading to zero. Add 5 drops phenolphthalein and 15 g. mannitol.

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Titrate with 0.5N NaOH through green to purplish blue. Add a heaping teaspoon of mannitol. If purple remains, the end point has been reached.

4.5.1.1.2.1 Calculation.

$$\text{Percent } Na_2B_8O_{13}{}^4H_2O = \frac{M \times N \times 0.05158 \times 100}{\text{Wt. sample}}$$

Where: M = ml. of NaOH.
N = normality of NaOH.

4.5.1.2 Determination of sodium chlorate.

4.5.1.2.1 Reagents.

Nitric acid, concentrated.
Sulphurous acid, 6 percent SO_2 solution.
0.1N silver nitrate. Standardize with NaCl that has been dried one hour at 100° C. and cooled in a desiccator over concentrated H_2SO_4.
0.1N ammonium thiocyanate.
Ferric nitrate solution. 407 g. of $Fe(NO_3)_3 9H_2O$ made up to 500 ml. with water.

4.5.1.2.2 Procedure. Weigh a 1.000 ± 0.001 g. sample and dissolve in 50 ml. water in a 300 ml. Erlenmeyer flask. Add 50 ml. sulfurous acid and boil until excess SO_2 is completely expelled, (indicated by absence of SO_2 odor), about 10 to 15 minutes. Cool the solution to room temperature, add 7 ml. concentrated HNO_3, 10 ml. ferric nitrate solution and then from a buret approximately 0.1 ml. 0.1N ammonium thiocyanate. Titrate with 0.1N $AgNO_3$ to a completely colorless endpoint. The net 0.1N $AgNO_3$ required for the chloride titration is obtained by deducting the number of ml. of 0.1N ammonium thiocyanate used from the total ml. 0.1N $AgNO_3$ used.

4.5.1.2.2.1 Calculation.

$$\text{Percent } NaClO_3 = \frac{M \times N \times 0.1064 \times 100}{\text{Wt. sample}}$$

Where: M = ml. $AgNO_3$.
N = normality of $AgNO_3$.

4.5.1.3 Inert ingredients. The percent inert ingredients is determined by subtracting the percent disodium octaborate tetrahydrate found plus the percent sodium chlorate found from 100 percent.

4.5.2 Water insoluble material.

4.5.2.1 Equipment.

Porcelain Gooch crucible, Coors No. 3A or equivalent.
Crucible holders.
Vacuum filter flask, 1000 ml. capacity.
Glass fiber filter paper, Hurlburt Paper Company.
No. X-934 size A or equivalent.

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4.5.2.2 Procedure. Prepare the filter by placing a glass filter pad in the Gooch crucible and wash it thoroughly with hot distilled water on the suction filter assembly. Dry in a 1050 C. oven for two hours, cool in a desiccator, and weigh. Dissolve 50 ± 0.1 g. sample in 200 ml. of 70° F. distilled water, let stand for one hour, then filter sample through the previously-prepared and weighed crucible on the suction assembly. Wash with 70°F. distilled water until completely free of soluble material. Dry for two hours in a 105° C. oven, cool in a desiccator, and reweigh.

4.5.2.3 Calculation.

Percent insoluble material = (wt. crucible and insoluble - tare wt. of crucible) X 2 X 100.

4.5.3 pH. Dissolve 24 ± 0.1 g. sample in 100 ml. of recently boiled and cooled distilled water. Determine the pH at 23° C. Any electric pH meter that will read to 0.1 pH point may be used for this test.

4.5.4 Appearance and pour. Measure 50 ml. of the dry herbicide into a glass cylinder seven inches high by 7/8 inch diameter, and equipped with a spout. By tilting the cylinder pour the herbicide at a uniform rate of approximately 100 ml. per minute. Observe the appearance of the herbicide in the cylinder and the pouring for conformance to 3.4.

4.6 Inspection of preparation for delivery. The packing and marking of the herbicide shall be inspected to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be Level A, B, or C, as specified (see 6.2).

5.1.1 Level A. The herbicide shall be packed in 50 ± 1/4 pound quantities in steel pails, in fiber drums, in bags, or in sacks. Steel pails shall conform to type II, class 4 of PPP-P-704 and shall be provided with a zinc coated steel bail. Each pail shall be provided with a close-fitting bag liner formed from plastic fiber of three mil minimum thickness conforming to type I, grade and finish optional of L-P-378. Bag liners shall be closed by tying, knotting, or heat sealing. Heat sealed seams shall meet the heat seal requirements of L-P-378. The filled pails shall be closed securely by crimpling the lug cover to the pail body. Fiber drums shall conform to PPP-D-723, type II, grade A. Bags shall conform to PPP-B-35, type II, class 2, 3, or 6. Sacks shall conform to UU-S-48 (X series).

5.1.2 Level B. The herbicide shall be packed in 50 ± 1/4 pound quantities in fiber drums conforming to PPP-D-723, type I, in bags conforming to PPP-B-35, type III, or in sacks conforming to UU-S-48 (L/W-series).

5.1.3 Level C. The herbicide shall be apcked in a manner which will insure arrival at first domestic destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules, or with rules and regulations of other carriers as applicable to the mode of transportation.

5.2 Marking.

5.2.1 The labeling shall be registered under the Federal Insecticide, Fungicide and Rodenticide Act of 1947, as amended in compliance with its requirements. The label data shall be durably and legibly marked in accordance with MIL-STD-129 on the panel of the body by lithographing, stenciling, or printing with a yellow, noncorrosive, weather-resistant, waterproof ink conforming to TT-I-558. Labels shall be securely affixed in place with water-resistant label adhesive as specified in MIL-A-3941 and shall be waterproofed by coating the entire surface of the label with the same adhesive.

5.2.2 Civil agencies. In addition to labeling specified in 5.2.1 and any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

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5.2.3 Military agencies. In addition to labeling specified in 5.2.1 and any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. CBM, chlorate-borate mixture is intended for use as a nonselective type weed killer and should be applied only in places where complete elimination of plant life is desired. It can be used to completely eliminate vegetation around antenna installations, railroad right-of-ways, roadsides and utility lines.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Inspection responsibility, methods, and tests (see sec. 4).
(c) Selections of applicable level of packing required (see 5.1).
(d) Type of package required.

6.3 The Federal Insecticide, Fungicide and Rodenticide Act of 1947 is under the jurisdication of the Pesticides Regulation Section, Plant Pest Control Branch, Agricultural Research Service, U. S. Department of Agriculture, Washington, D. C., 20250 to which agency all matters pertaining to regulation should be addressed.

MILITARY INTEREST:

Custodians:

Army - MU
Navy - YD
Air force - 68

Review activities:

Army - MU
Navy - YD
Air force - 68

Preparing activity:

Air force - 68

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price . cents each.

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